

How do we find the projection of a given vector $\mathbf{v}$ onto a subspace V ?

Answer:

Finding the projection $\mathbf{p}$ of a vector $\mathbf{u}$ onto a subspace V means finding the vector in V that is "closest" to $\mathbf{u}$. This projection $\mathbf{p}$ can be thought of as the "shadow" that $\mathbf{u}$ casts onto the subspace V . (Note that we usually assume $\mathbf{u}$ does not belong to V . Otherwise the projection of $\mathbf{u}$ is just $\mathbf{u}$ itself.)

There are several methods to find the projection. In our course MA1513, we use the least squares approach as follow:

1. **Find a basis for the subspace V :** Depending on the type of subspace V is given, we can find a basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ for V accordingly.
2. **Form the Matrix $\mathbf{A}$:** Form a matrix $\mathbf{A}$ whose columns are the basis vectors in step 1.

$$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_k \\ | & | & & | \end{bmatrix}$$

The subspace V is now the column space of this matrix $\mathbf{A}$.

3. **Find the Least Squares Solution:** To find the least squares solution of $\mathbf{Ax} = \mathbf{u}$, we solve the "normal equation": $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{u}$.

The coefficient matrix $\mathbf{A}^T \mathbf{A}$ is guaranteed to be non-singular as long as the columns of $\mathbf{A}$ are linearly independent, which they are since they form a basis.

So the least squares solution is given by $\mathbf{x}_0 = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{u}$.

4. **Calculate the Projection Vector:** The projection vector $\mathbf{p}$ is obtained by pre-multiplying the least squares solution by $\mathbf{A}$: $\mathbf{p} = \mathbf{Ax}_0$.

Note that the right hand side is the linear combination of the columns of $\mathbf{A}$ with the coefficients from $\mathbf{x}_0$.

Substituting the expression for $\mathbf{x}_0$ from the previous step gives the formula:

$$\mathbf{p} = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{u}$$

The matrix $\mathbf{M} = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ in the above formula is called the **projection matrix**. Once you compute $\mathbf{M}$, you can find the projection of *any* vector $\mathbf{v}$ onto the subspace V by simply multiplying $\mathbf{v}$ by this projection matrix $\mathbf{M}$.

Note: If the subspace V is the column space of a given matrix $\mathbf{A}$, we cannot assume all the columns of $\mathbf{A}$ are linearly independent (there may be "redundant" vectors among the columns). In this case, $\mathbf{A}^T \mathbf{A}$ is singular and the formula in step 4 fails. To overcome this, you should identify the redundant columns in $\mathbf{A}$ (by looking at the non-pivot columns of the REF/RREF of $\mathbf{A}$). Remove these redundant columns from $\mathbf{A}$ to get another matrix $\mathbf{A}_1$. Use this new matrix $\mathbf{A}_1$ to find the projection:

$$\mathbf{p} = \mathbf{A}_1(\mathbf{A}_1^T \mathbf{A}_1)^{-1} \mathbf{A}_1^T \mathbf{u}$$

Similarly, suppose the subspace V is a linear span of a set of vectors $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. Instead of using the entire spanning set to form a matrix $\mathbf{A}$, we need to remove the redundant vectors from the set first before forming the matrix $\mathbf{A}$ and apply the formula in step 4 to find the projection.

Scenario 1

Let V be a plane in the 3D space with equation $2x - 2y + z = 0$. Find the projection of the vector $\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$ onto the plane V .

1. **Find a basis for V :** We just need to choose two non-parallel vectors that lie on the plane. This can be easily done by finding any value for x , y , and z that satisfy the equation $2x - 2y + z = 0$. For example, $x = 1$, $y = 1$ and $z = 0$ which gives a vector $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$; and $x = 0$, $y = 1$ and $z = 2$ which gives another vector $\mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$. The plane V is the set of all linear combinations of these two vectors (i.e., the span of $\{\mathbf{a}_1, \mathbf{a}_2\}$).

2. **Form the matrix A :** The columns of A are our basis vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 2 \end{bmatrix}$

3. **Find the least squares solution of $A\mathbf{x} = \mathbf{u}$:** $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$.

We solve the normal equation $A^T A \mathbf{x} = A^T \mathbf{u}$: $\begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$.

This gives the least squares solution $\mathbf{x}_0 = (A^T A)^{-1} A^T \mathbf{u} = \frac{1}{9} \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{17}{9} \\ \frac{2}{9} \end{bmatrix}$

4. **Find the projection $\mathbf{p}$:** $\mathbf{p} = A\mathbf{x}_0 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{17}{9} \\ \frac{2}{9} \end{bmatrix} = \begin{bmatrix} \frac{17}{9} \\ \frac{19}{9} \\ \frac{4}{9} \end{bmatrix}$

Scenario 2

Let V be the column space of the matrix $A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{bmatrix}$. Find the projection of the vector $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ onto the subspace V .

A 3×4 matrix has 4 column vectors in $\mathbb{R}^3$. Since the dimension of the space is 3, any set of 4 vectors must be linearly dependent. This means the matrix $A^T A$ will be singular, and we cannot use the standard formula $A(A^T A)^{-1} A^T \mathbf{u}$ directly on A .

The correct approach is to first find a basis for the column space of A , and then use that basis to form a new matrix to find the projection.

1: Find a basis for the column space V

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{bmatrix} \xrightarrow{GJE} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns in the RREF are columns 1 and 2. Therefore, a basis for the column space V is formed by the first and second columns of the **original matrix $\mathbf{A}$** .

Our basis for V is $\{\mathbf{b}_1, \mathbf{b}_2\}$, where: $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

2: Form the matrix $\mathbf{B}$. Using the two basis vectors $\{\mathbf{b}_1, \mathbf{b}_2\}$, we form the 3×2 matrix $\mathbf{B} =$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

The column space of $\mathbf{B}$ is the same as the column space of $\mathbf{A}$, which is our subspace V .

3: Find the least squares solution of $\mathbf{Bx} = \mathbf{u}$: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

We solve the normal equation $\mathbf{B}^T \mathbf{Bx} = \mathbf{B}^T \mathbf{u}$: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

This gives the least squares solution $\mathbf{x}_0 = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \mathbf{u} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}$

4: Find the projection $\mathbf{p}$: $\mathbf{p} = \mathbf{Bx}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{4}{3} \end{bmatrix}$

Note: If you use the original matrix $\mathbf{A}$ in step 2 and find the least squares solution of $\mathbf{Ax} = \mathbf{u}$ in step 3, you will have to use GE or GJE to solve the normal equation $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{u}$ instead of the formula, since $\mathbf{A}^T \mathbf{A}$ is singular. You should expect infinitely many solutions, meaning there are infinitely many choices of least squares solutions $\mathbf{x}_0$. You may use any one of the specific least squares solutions $\mathbf{x}_0$ to find the projection by pre-multiplying the original matrix $\mathbf{A}$ to $\mathbf{x}_0$.

Here is the alternative process using the original matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{bmatrix}$

To find the least squares solutions, we solve the **normal equation $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{u}$**

$$\begin{bmatrix} 2 & 1 & 3 & 3 \\ 1 & 2 & 3 & 0 \\ 3 & 3 & 6 & 3 \\ 3 & 0 & 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 2 \end{bmatrix}$$

The coefficient matrix is singular. So we solve the system by applying GJE to the augmented matrix:

$$\begin{bmatrix} 2 & 1 & 3 & 3 & 2 \\ 1 & 2 & 3 & 0 & 2 \\ 3 & 3 & 6 & 3 & 4 \\ 3 & 0 & 3 & 6 & 2 \end{bmatrix} \xrightarrow{GJE} \begin{bmatrix} 1 & 0 & 1 & 2 & \frac{2}{3} \\ 0 & 1 & 1 & -1 & \frac{2}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The general solution of the normal equation is given by

$$\mathbf{x}_0 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} - s - 2t \\ \frac{2}{3} - s + t \\ s \\ t \end{bmatrix} \text{ where } s \text{ and } t \text{ can be any real numbers.}$$

There are infinitely many least squares solutions. Each vector $\mathbf{x}_0$ in this set will produce the exact same projection when multiplied by $\mathbf{A}$.

Let's verify this with the simplest solution (when $s=0$ and $t=0$): $\mathbf{x}_0 = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ 0 \\ 0 \end{bmatrix}$

$$\text{The projection is } \mathbf{Ax}_0 = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{4}{3} \end{bmatrix}$$

This is exactly the projection vector $\mathbf{p}$ we found above.

Note that any choice of s and t for $\mathbf{x}_0$ will result in this same projection vector, which is the closest vector in the column space of $\mathbf{A}$ to $\mathbf{u}$.